Toolkit: Chemistry of the Earth

Practical Ten: Radioisotopes and Geochronology

Objectives:

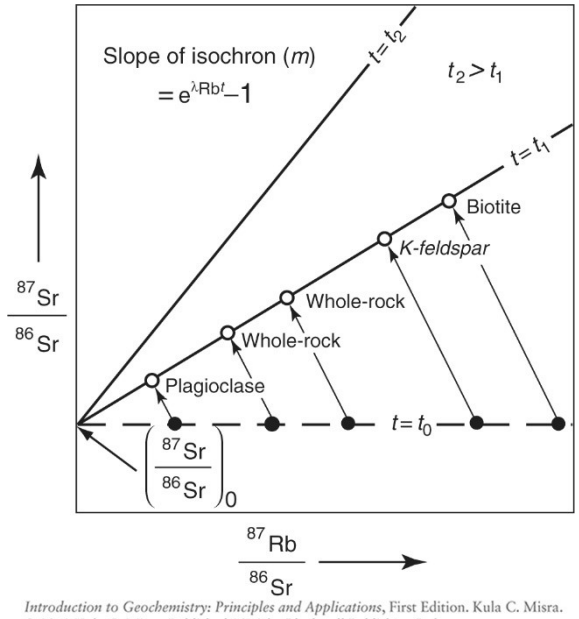
- Understand the principles of radioactive decay
- Investigate how isochrons are used to determine the ages of rocks

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FEEDBACK

$$D = D_0 + N (e^{\lambda t} - 1)$$

$$\left[\frac{^{87}Sr}{^{86}Sr}\right]_m = \left[\frac{^{87}Sr}{^{86}Sr}\right]_0 + \left[\frac{^{87}Rb}{^{86}Sr}\right]_m (e^{\lambda t} - 1)$$



Introduction to Geochemistry: Principles and Applications, First Edition. Kula C. Misra.
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A. Radioactive decay and half-lives

We are going to consider the decay of radioactive parent ^{87}Rb and the growth of daughter isotope ^{87}Sr in a granite sample, over the course of six half-lives. We will assume that the granite initially contains 1.2×10^{20} atoms of ^{87}Rb and 0.3×10^{20} atoms of ^{87}Sr. In other words,

$N_0 = 1.2 \times 10^{20}$ and $D_0 = 0.3 \times 10^{20}$

1. The half-life of ^{87}Rb is 48.8×10^9 years. Calculate the decay constant λ, using the equation:

$$\lambda = 0.693 / t_{½}$$

$\lambda = 1.4201 \times 10^{-11}$

2. Use the equation, $N/N_0 = e^{-\lambda t}$

to determine the number of atoms of ^{87}Rb (N) after one half-life.

$N = N_o e^{-\lambda t}$

$N = (1.2 \times 10^{20}) * (\exp((-1.4201 \times 10^{-11}) * (48.8 \times 10^9)))$

$N = 6.00 \times 10^{19}$

Of course, the value of N after 1 half-life is half the number of atoms to begin with (i.e. $N = N_0/2$). However, it is satisfying to see that the equation works and gives the correct answer.

3. Calculate the number of daughter atoms of ^{87}Sr present after one half-life, using the relationships,

$D^* = N_0 - N$ and $D = D_0 + D^*$

where D* is the amount of radiogenic ^{87}Sr, produced from decay of ^{87}Rb, after the rock formed.

$D^* = N_0 - N = (1.2 \times 10^{20}) - (6.00 \times 10^{19}) = 6.00 \times 10^{19}$

- again, this makes sense. After one half-life, half the parent atoms have converted to daughter atoms, so the values are the same.

$D = D_0 + D^* = (0.3 \times 10^{20}) + (6.00 \times 10^{19}) = 9.00 \times 10^{19}$

Your calculations should agree with the first two rows of the table below:

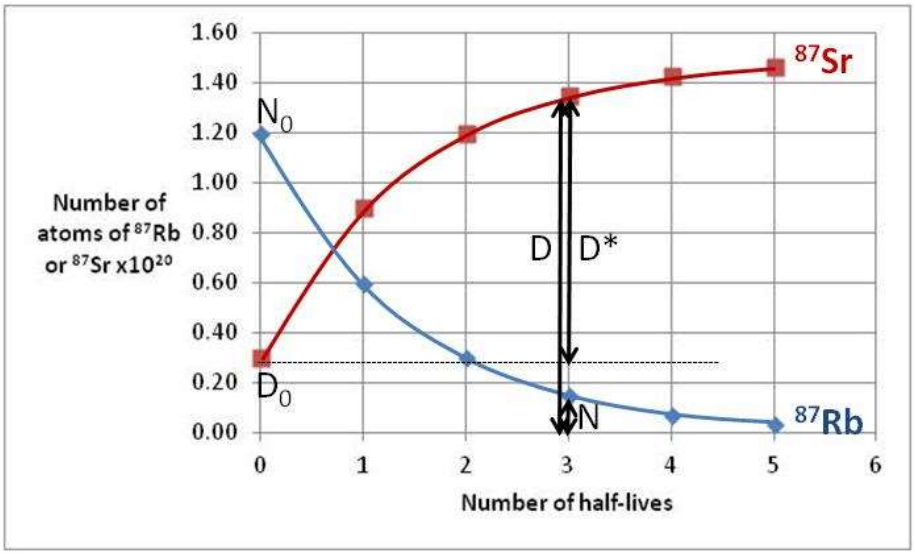
Table 1: decay of ^{87}Rb			
Number of half-lives	N	D*	D
0	1.20E+20	0.00E+00	3.00E+19
1	6.00E+19	6.00E+19	9.00E+19
2	3.00E+19	9.00E+19	1.20E+20
3	1.50E+19	1.05E+20	1.35E+20
4	7.50E+18	1.12E+20	1.42E+20
5	3.75E+18	1.16E+20	1.46E+20

For the third row of the table, $N = (1.2 \times 10^{20}) * (\exp((-1.4201 \times 10^{-11}) * 2 * (48.8 \times 10^9)))$ etc….

4. The same values from the Table 1 are shown in Table 2, calculated with the same factor, x 10^{20}:

Table 2: decay of ^{87}Rb - values for graph			
Number of half-lives	N x 10^{20}	D* x 10^{20}	D x 10^{20}
0	1.20	0.00	0.30
1	0.60	0.60	0.90
2	0.30	0.90	1.20
3	0.15	1.05	1.35
4	0.08	1.12	1.42
5	0.04	1.16	1.46

Use the data in Table 2 to plot a graph showing the decay of ^{87}Rb and the growth of ^{87}Sr in the granite. Plot two curves: N and D vs. Number of half-lives.



On the graph, add labels to show the meaning of N, N_0, D, D* and D_0, for the case when the number of half-lives = 3.

(Nb. This exercise is based on the worked example, 14.2, in McSween (page 289). In this example, the figure and figure caption for Fig. 14.2 have important errors. The figure caption says that the graph shows decay of ^{87}Sr to ^{87}Rb – this is the wrong way round. The text is correct, the figure caption is not. Also, on the figure the decaying curve is labelled N_{Sr}. This is incorrect – it should read N_{Rb}. This is an example of how you should always read your textbooks critically.)

B. Determining the age of a rock from an isochron

(Taken from McSween, Worked example 14.3, page 293)

Table 3: Isotope data for Apollo basalt				
Mineral	Rb (ppm)	Sr (ppm)	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr$
Whole rock	0.926	90.4	0.02960	0.70096
Plagioclase	0.599	323	0.00537	0.69989
Pyroxene	0.386	22.7	0.04920	0.70200
Ilmenite	3.76	96.5	0.11270	0.70490

Table 3 shows Rb-Sr data measured for an Apollo 12 lunar basalt (sample 12014) by L. Nyquist and co-workers. (Nyquist et al. (1979) Proc. Lunar Planet Sci Conf 10, 77-114)

To see images of this rock, including an interactive Virtual Microscope, go to:

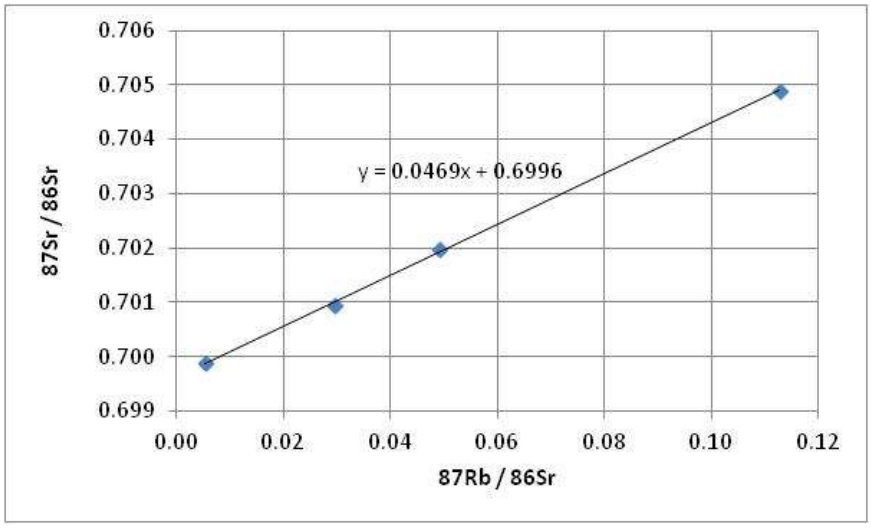
http://www.virtualmicroscope.org/content/12014-5-olivine-basalt

The **isochron equation** for the Rb-Sr system is:

$$\left[\frac{^{87}Sr}{^{86}Sr}\right]_m = \left[\frac{^{87}Sr}{^{86}Sr}\right]_0 + \left[\frac{^{87}Rb}{^{86}Sr}\right]_m (e^{\lambda t} - 1)$$

What is the age of the rock?

1. To answer this question, we first need to draw the isochron. Plot the appropriate data on the graph below and label the axes.



2. Draw the best fit line to the data, and determine the value of the intercept on the y axis:

This graph is drawn in Excel. The equation of the line is calculated using a linear regression. Your value of the intercept should be about 0.6996.

What part of the isochron equation does the intercept represent?

$(^{87}Sr / ^{86}Sr)_0$, i.e. the initial ratio of ^{87}Sr to ^{86}Sr when the rock formed.

3. To use the geochronology equation, we need the slope of the line. Choose two data points (minerals, whole-rock and/or the intercept) to determine the slope $(y_2-y_1)/(x_2-x_1)$. Explain why you chose the two points you selected.

You might choose for example, ilmenite and plagioclase, since they are at the ends of the line. In this case, the slope m is equal to:

$$m = (0.70490 - 0.69989) / (0.11270 - 0.00537) = 0.04668$$

This is close to, but not exactly the same as, the slope determined in the linear regression shown above.

4. The slope of the line (m) is equal to $(e^{\lambda t}-1)$:

$$m = (e^{\lambda t} - 1)$$

Rearrange the equation to determine t, the time since the rock formed.

$$m = (e^{\lambda t} - 1)$$

$$(m + 1) = e^{\lambda t}$$

$$\ln (m + 1) = \lambda t$$

$$(\ln (m + 1)) / \lambda = t$$

Then calculate t, the age of the rock. Don't forget to include units.

$$t = (\ln (0.04668 + 1)) / (1.4201 \times 10^{-11})$$

$$t = 3.21 \times 10^{9} \text{ years} = 3.21 \text{ billion years}$$

5. Compare the age you determined to ages that other students calculated using different pairs of minerals etc. to determine the slope. Comment on the differences between the calculated ages.

The age calculated using the slope calculated in the linear regression is 3.23 billion years. Calculating the slope of the isochron "by hand" the way we just did it is bound to be the least accurate method. However, it is not far from the mathematically best answer.

The uncertainty on the age of the rock determined by Nyquist et al. is ±0.11 billion years, based on fitting a linear regression to the data. Where do you think errors and uncertainties might arise?

Uncertainties can arise because of small deviations of one point from the isochron. The more points there are to define the isochron, the better the certainty of the fit. Errors in the data can arise from analytical uncertainties. Think about how precisely we need to know the values of the isotope ratios – we are looking at the fourth decimal place for $^{87}Sr / ^{86}Sr$ (at least). Measuring these ratios requires a very good mass spectrometer. It is essential that the mineral grains analysed are not contaminated, for example by secondary minerals that have a younger age. It is also possible to contaminate the analysis if there is a tiny amount of Rb or Sr in the acids used to digest (dissolve) the mineral grains. Radiometric dating requires very precise analytical skill!